Phase 1: Structured Diagnosis

Case A: Our World in Data COVID-19 Weekly Cases Map

The OWID map looks clear and authoritative at first glance, but its default settings nudge viewers to compare countries using raw totals rather than a more meaningful, burden-based perspective. The screenshot below highlights the interface cues that subtly guide viewers toward this interpretation.

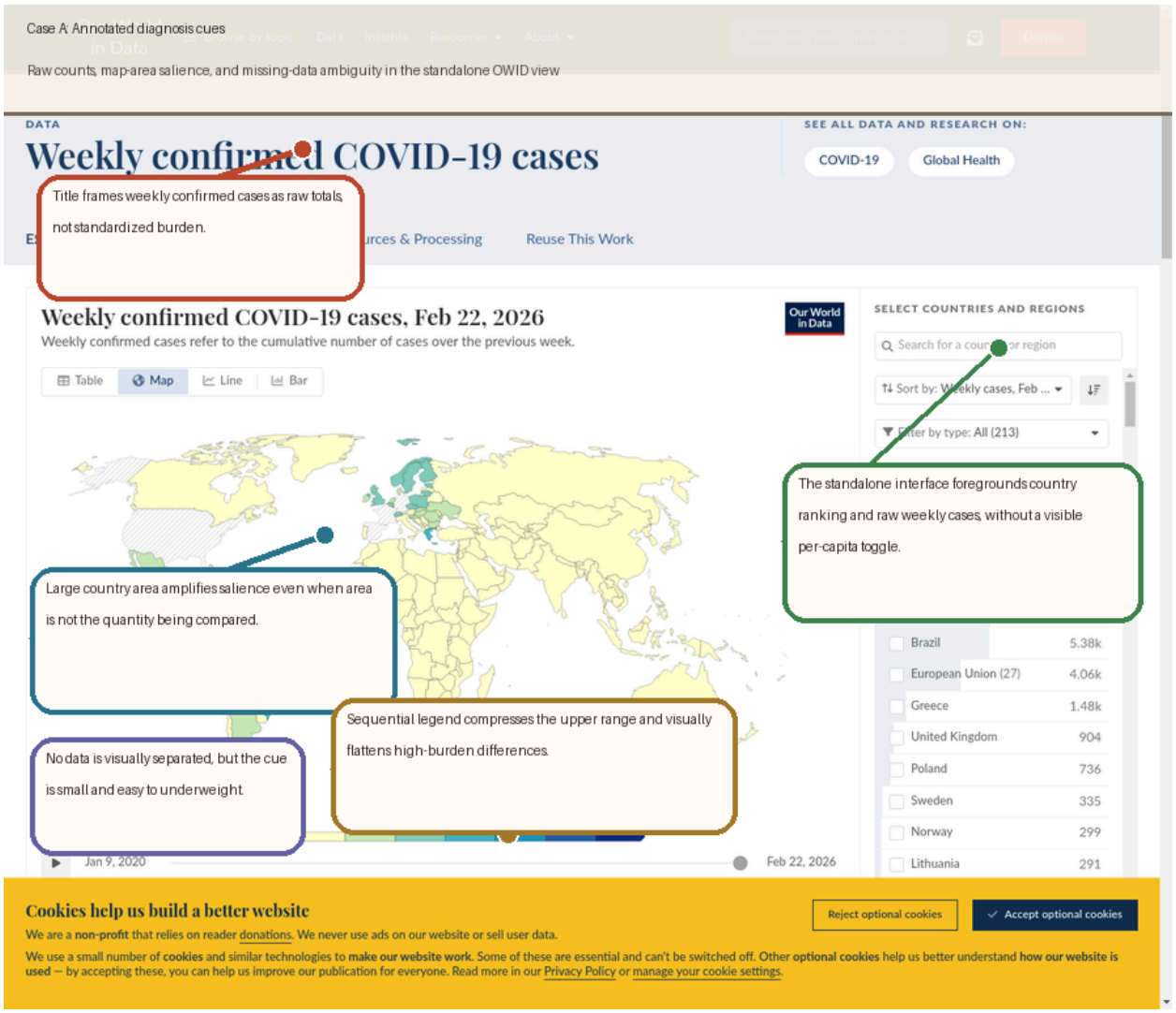

Figure 1. *Annotated Case A screenshot showing the raw-count title framing, the choropleth map, the compressed sequential legend, the small no-data cue, and the country ranking panel.*

Table 1: COVID-19 Visualization Bias Analysis

Where it appears	Bias and misleading impression	Severity	Root cause	Course-theory grounding
The title and legend foreground weekly confirmed COVID-19 cases in raw counts, and the standalone view does not visibly foreground a relative-to-population alternative.	Absolute counts are framed as the default basis for comparison. This can make viewers believe that countries with larger totals are necessarily experiencing the more serious outbreak.	**Dangerous.** In a public-health context, a reasonable decision-maker could underweight a smaller country with a severe per-capita outbreak because the default presentation makes raw totals appear to be the correct comparison basis.	Create or convert; Principle of Consistency; Deceptive Visualization	Framing + interaction design
The choropleth gives geographically large countries far more screen area than small countries.	Country area becomes a visual proxy for importance even though land area is not the variable under analysis.	**Misleading.** A casual viewer can over-attend to large countries simply because they occupy more visual space, even if the intended comparison concerns disease burden rather than territorial footprint.	Choosing Visual Encodings; Visual Effectiveness Rankings	Visual encoding

Upper-end legend compression weakens comparison among high values	The sequential color scale culminates in a dark upper band up to about 1 million, which visually flattens differences among high-burden countries. This can leave the impression that all dark countries are roughly equivalent.	**Misleading.** The chart still identifies which countries are high, but it weakens discrimination where comparison is often most important.	Poor Scaling; Information in Hue and Lightness	Visual encoding
Missingness and measurement comparability are visually under-signaled	The legend includes a "No data" swatch, but it is visually minor, and the map does not strongly encode uncertainty arising from uneven testing and reporting practices. This can make non-reporting or weak comparability feel like low severity.	**Misleading.** The distortion may not directly drive a policy decision by itself, but it can give viewers a stronger sense of certainty and comparability than the data justify.	Data Examination; Critique your prototype	Data + visual encoding

The most significant issue in Case A is the raw-count default, as it shapes the viewer's first and strongest interpretation before they can question whether the comparison is appropriate. The remaining issues reinforce this framing by making the map appear more spatially and statistically straightforward than the underlying data actually are.

Case B: Gapminder Bubble Chart

Gapminder is compelling because it makes long-term development trends easy to remember, but its default presentation is also rhetorically powerful. It encourages viewers to watch countries move, infer smooth historical progress from that motion, and accept the logarithmic income axis as a natural choice rather than a meaningful design decision.

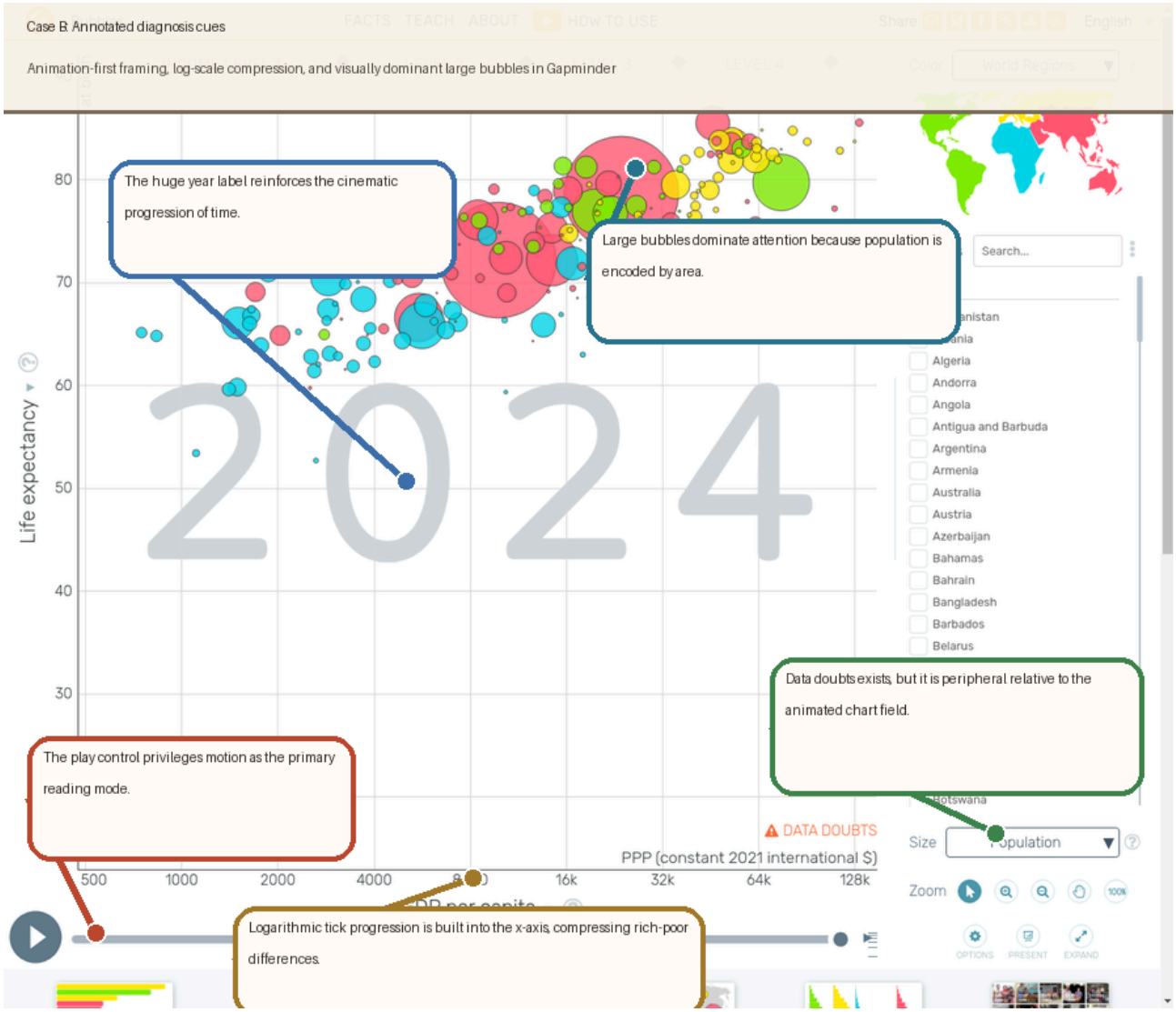

***Figure 2**. Annotated Case B screenshot showing the play control, oversized year staging, logarithmic income axis, visually dominant large bubbles, and peripheral uncertainty cue.*

Table 2: Interactive Animation Bias Analysis

Where it appears	Bias and misleading impression	Severity	Root cause	Course-theory grounding
The play control and the large year overlay make temporal motion the primary reading mode.	Animation encourages a smooth progress narrative. As bubbles drift upward and rightward, viewers are encouraged to infer coherent developmental progress and sometimes even implied causality.	Misleading. The chart does not falsify the data, but it strongly nudges viewers toward a persuasive story that is more coherent than the evidence alone warrants.	Constructing Narratives; Attribution of Causality; Problems of Animation	Interaction design + framing
During playback, many bubbles move simultaneously while also varying in size and color.	The interface exceeds human multiple-object tracking limits. Most viewers cannot accurately track more than a small subset of them, so the display pushes them toward that many impressionistic reading.	Misleading. The interface feels analytically precise, but much of that apparent precision is lost because the viewer cannot actually follow that many moving marks at once.	Multiple Object Tracking; When Does Animation Help?	Interaction design
The default log-scale x-axis compresses	The income axis uses logarithmic spacing, visible in the tick progression	Misleading. A log scale can be analytically defensible, but when it is	Poor Scaling; Mapping Data to Visual Variables;	Visual encoding + framing

rich-poor differences	from 500 to 128k. This compresses high-end income differences and can make countries appear more convergent than they would on a linear scale.	hidden inside the default view it privileges one interpretation while obscuring the rhetorical effect of the choice.	Angle, Framing and Focus	
Playback and trails create fluid paths even though the dataset is sampled discretely and not all years carry equal certainty.	Smooth motion overstates continuity and certainty in the historical record. This can make historical development look continuously observed rather than sampled and partially regularized.	Misleading. The smoothness comes partly from the interface, not only from the data, so the display can overstate continuity in the evidence.	Congruence Principle; Animation vs. Small Multiples; Data Examination	Data + interaction design

The core issue in Case B is therefore not a single factor, but a combination of design choices. Animation creates narrative momentum, the log scale alters how comparisons are perceived, and the interface smooths historical change into continuous motion. Together, these elements make the visualization memorable and persuasive, but not neutral.

Diagnostic Conclusion

Across the two cases, the most severe bias in Case A is the emphasis on absolute weekly case counts in a public health comparison. In Case B, the key issue is the combination of animation-first storytelling and hidden scale assumptions. These findings support selecting Case B for redesign, as an interactive alternative can address both problems, by replacing autoplay with comparison-focused views and by making the impact of scale choice explicit

rather than hidden.

References

1. Assignment Brief

2. Our World in Data - Weekly confirmed COVID-19 cases. https://ourworldindata.org/grapher/weekly-covid-cases

3. Gapminder Tools. https://www.gapminder.org/tools/

4. Lecture 2, 5, 10 Slides